

HAR-SMI-500: COMPONENT & MATHEMATICAL REFERENCE MANUAL

Authoritative Engineering Specification & Derivations Across All Schematic Pages
500W High-Altitude Resilient Solar Micro-Inverter | DRDO | iDEX [SIH26049] | 5,000m HAA Ladakh Envelope

Executive System Overview: The **HAR-SMI-500** is an ultra-reliable, high-efficiency (peak 97.4%) solar micro-inverter engineered specifically for the extreme high-altitude environment of Ladakh (5,000m above sea level). Operating at an atmospheric pressure of 54.05 kPa (53% of sea level) and temperatures ranging from -40°C to +50°C, the converter addresses two primary high-altitude failure modes: (1) Dielectric breakdown due to reduced mean free path (Paschen's Law), and (2) Sub-zero cold starts causing gate driver latchup and thermal stress.

Global Environmental & System-Level Mathematical Formulations:

1. Barometric Pressure Model: $P(h) = P_0 * (1 - L * h / T_0)^{(g * M / R_0 * L)} = 101.325 * (1 - 0.0065 * 5000 / 288.15)^{5.2558} = \mathbf{54.05\text{ kPa}}$ (53.3% of 1 atm)
2. Paschen Clearance Derating: $k_d = (P_0 / P_{HAA})^{0.8} = (101.325 / 54.05)^{0.8} = \mathbf{1.480x}$ -> Min Clearance $d_{HAA} = 1.48 * 3.0\text{mm} \geq \mathbf{4.5\text{ mm}}$, Creepage $\geq \mathbf{6.0\text{ mm}}$
3. System Power & Efficiency: $P_{in(nom)} = 40.0V * 12.83A = 513.2W$, $P_{out} = 230.0V * 2.174A = 500.0W$ -> Efficiency = 500 / 513.2 = 97.43%
4. Thermal Dissipation Budget: $P_{loss} = P_{in} - P_{out} = 13.2W$ -> $R_{th(h-a)} = (T_{j(max)} - T_{amb}) / P_{loss} = (125\text{ C} - 50\text{ C}) / 13.2W = \mathbf{5.68\text{ C/W}}$
5. Dual-Stage Energy Buffer: $E_{stored} = (1/2) * C_{DC} * V_{DC}^2 = 0.5 * (18.8\text{ uF}) * (400V)^2 = \mathbf{1.504\text{ Joules}}$ -> Hold-up Time = 1.504J / 500W = 3.01 ms
6. Inverter Modulation Index: $M = (\sqrt{2} * V_{grid(rms)}) / V_{DC(bus)} = (\sqrt{2} * 230V) / 400V = 325.27V / 400.0V = \mathbf{0.8132}$ (Linear SPWM region)

Hierarchical Schematic Sheet Structure (1-to-1 Mapping):

Page	Schematic Sheet File	Subsystem Function	Key Power / Control Topology
Page 1	HAR_SMI_500.kicad_sch	Master Root Navigator	System Hierarchy, Overview & Global Derating Parameters
Page 2	01_PV_Input_PTC.kicad_sch	PV Input & Thermal Conditioning	20V-60V DC, 20A Fuse, Schottky, TVS, CM Choke, 25W PTC Heater
Page 3	02_LLC_Primary_GaN.kicad_sch	GaN LLC Resonant Converter	4x GS66508B GaN Full-Bridge, UCC21520 Drivers, 100kHz Planar T1
Page 4	03_Secondary_Rectifier_DCLink.kicad_sch	Secondary Rectifier & 400V DC Link	4x C3D04060A SiC Schottky, RC Snubber, 18.8uF Film DC Link, Divider
Page 5	04_SiC_Inverter_LCL_Grid.kicad_sch	SiC Inverter & Grid Interface	4x C3M0075120K SiC MOSFETs (Kelvin Source), LCL Filter, Relays
Page 6	05_Control_DualMCU_Sensors.kicad_sch	Dual Isolated MCU & Sensors	TM4C1230 (DSP) + ESP32-S3 (Comms) + ISO7762 + BME280 HAA Sensor
Page 7	06_Aux_Power_Supply.kicad_sch	Multi-Rail Auxiliary Supply	UCC3800 Flyback, 3-Winding T601, +15V_PRI/SEC, +5V, +3.3V Rails

Sheet 1 (Page 2): 01_PV_Input_PTC — PV Input Stage, Filtering & Pre-Thermal Conditioning

Accepts DC power from 20V to 60V (nominal 40V, open-circuit max 65V). Incorporates reverse polarity protection, TVS surge clamping, differential/common-mode EMI filtering, current monitoring, and a 25W autonomous PTC heater warming the core from -40°C.

Key Engineering Equations & Altitude Deratings:

* Paschen Derating: $k_d = (P_0 / P_{HAA})^{0.8} = (101.325 / 54.05)^{0.8} = \mathbf{1.48x}$ -> Min Clearance $\geq \mathbf{4.5\text{ mm}}$, Creepage $\geq \mathbf{6.0\text{ mm}}$
* PTC Pre-Heating: $P_{PTC} = V_{PV}^2 / R_{PTC} = (40V)^2 / 64\text{ Ohm} = \mathbf{25.0\text{ W}}$ -> $\Delta T = (P_{PTC} \cdot t) / (m \cdot c_p)$ approx 40 C in 3.2 min (warms from -40 C)
* Input Pi-Filter: $f_c = 1 / (2 \cdot \pi \cdot \sqrt{L_{101} \cdot C_{in}}) = 1 / (2 \cdot \pi \cdot \sqrt{0.8mH \cdot 480\mu F}) = \mathbf{256.7\text{ Hz}}$ ($C_{in} = C_{101} + C_{102} = 10\mu F + 470\mu F$)
* Current Shunt: $V_{CS} = I_{PV} \cdot R_{101} \cdot G_{U101} = I_{PV} \cdot 2.0\text{ mOhm} \cdot 20\text{ V/V} = \mathbf{40.0\text{ mV/A}}$ -> 12.5 A (full load) = 0.500 V to ADC

Complete Component Bill of Materials & Electrical Ratings:

Ref	Footprint / Library Symbol	Description	Electrical Ratings & Values	Circuit Function & Role
J101	TerminalBlock_Phoenix_MKDS-1,5-2	PV Input Connector	20-60V DC, 20A rated	5.08mm pitch high-current terminal block
F101	Fuse_Blade_ATO_directSolder	Fast-Blow Blade Fuse	20A, 80V DC rating	Primary overcurrent / short-circuit isolation
D101	TO-220-2_Verical (V30100P)	Schottky Barrier Diode	100V, 30A, Vf=0.48V	Reverse PV polarity blocking diode
D102	Diode_SMD:D_SMC (5.0SMDJ58A)	TVS Diode 5.0SMDJ	58V Vwm, 93.6V Vclamp, 5kW	Input surge & lightning overvoltage clamp
L101	Inductor_SMD:L_12x12mm	Common-Mode Choke	0.8 mH, 15A DC saturation	Attenuates common-mode conducted EMI
C101	Capacitor_SMD:C_0805	High-Freq Ceramic Cap	10 uF, 100V X7R dielectric	High-frequency input ripple bypass
C102	Capacitor_THT:CP_Radial_D12.5mm	Electrolytic Bulk Cap	470 uF, 100V, low-ESR, 105C	Bulk energy buffer & low-frequency filtering
R101	Resistor_SMD:R_2512	Kelvin Current Shunt	2.0 mOhm, 3W, 1% tol	Primary PV input current sensing
R_PTC	Resistor_SMD:R_2512	PTC Heating Element	64 Ohm, 35W pulse rated	Core warmup from -40C cold-start
D103	Diode_SMD:D_SOD-123	Freewheeling Diode	100V, 300mA, trr=4ns	PTC heater inductive clamp diode
Q101	Package_TO_SOT_SMD:SOT-23	NMOS Control Switch	30V, 5.7A, Rds(on)=18mOhm	Low-side switch controlling PTC warmup
Rg101	Resistor_SMD:R_0805	Gate Resistor	100 Ohm, 0.125W	Gate damping resistor for Q101
U101	Package_SO:SOIC-8	Current Sense Amplifier	Gain=20 V/V, 80V CM rejection	Precision bi-directional current monitor
C103	Capacitor_SMD:C_0805	Decoupling Capacitor	100 nF, 50V X7R	Power supply local decoupling for U101

Sheet 2 (Page 3): 02_LLC_Primary_GaN — Full-Bridge GaN LLC Resonant DC-DC Converter

Converts variable 20V-60V DC into a high-voltage resonant AC waveform. Features 4x GS66508B GaN-on-Si HEMTs in full-bridge topology driven by 2x UCC21520 isolated gate drivers, zero-voltage switching (ZVS), and a custom planar EILP38 1:8 transformer.

Key Engineering Equations & Altitude Deratings:

* LLC Resonant Freq: $f_r = 1 / (2\pi \cdot \sqrt{L_r \cdot C_r}) = 1 / (2\pi \cdot \sqrt{15.0\text{ uH} \cdot 169\text{ nF}}) = \mathbf{100.0\text{ kHz}}$
* Inductance Ratio: $k = L_m / L_r = 75.0\text{ uH} / 15.0\text{ uH} = \mathbf{5.0}$ (Ensures wide ZVS zero-voltage switching operational range)
* Characteristic Z: $Z_0 = \sqrt{L_r / C_r} = \sqrt{15.0\text{ uH} / 169\text{ nF}} = \mathbf{9.42\text{ Ohm}}$, Quality Factor $Q = Z_0 / R_{ac} = 9.42 / 8.10 = \mathbf{1.16}$
* Transformer Ratio: $n = N_s / N_p = 16 / 2 = \mathbf{8.0}$ -> $V_{sec} = 8.0 \cdot 40V = \mathbf{320\text{ V}}$ (Nominal secondary peak)

Complete Component Bill of Materials & Electrical Ratings:

Ref	Footprint / Library Symbol	Description	Electrical Ratings & Values	Circuit Function & Role
Q201	Package_TO_SOT_SMD:SOT-23 (GS66508B)	GaN HEMT Transistor	650V, 30A, Rds=50mOhm	Full-Bridge Leg A High-Side GaN Switch
Q202	Package_TO_SOT_SMD:SOT-23 (GS66508B)	GaN HEMT Transistor	650V, 30A, Rds=50mOhm	Full-Bridge Leg A Low-Side GaN Switch
Q203	Package_TO_SOT_SMD:SOT-23 (GS66508B)	GaN HEMT Transistor	650V, 30A, Rds=50mOhm	Full-Bridge Leg B High-Side GaN Switch
Q204	Package_TO_SOT_SMD:SOT-23 (GS66508B)	GaN HEMT Transistor	650V, 30A, Rds=50mOhm	Full-Bridge Leg B Low-Side GaN Switch
Rg1	Resistor_SMD:R_0805	GaN Gate Resistor	4.7 Ohm, 0.125W 1% tol	Limits dV/dt & gate turn-on ringing for Q201
Rg2	Resistor_SMD:R_0805	GaN Gate Resistor	4.7 Ohm, 0.125W 1% tol	Limits dV/dt & gate turn-on ringing for Q202
Rg3	Resistor_SMD:R_0805	GaN Gate Resistor	4.7 Ohm, 0.125W 1% tol	Limits dV/dt & gate turn-on ringing for Q203
Rg4	Resistor_SMD:R_0805	GaN Gate Resistor	4.7 Ohm, 0.125W 1% tol	Limits dV/dt & gate turn-on ringing for Q204
U201	Package_SO:SOIC-16W (UCC21520DW)	Isolated Dual Gate Driver	5.7 kVrms isolation, 4A/6A drive	Reinforced isolated gate driver for Leg A
U202	Package_SO:SOIC-16W (UCC21520DW)	Isolated Dual Gate Driver	5.7 kVrms isolation, 4A/6A drive	Reinforced isolated gate driver for Leg B
Lr	Inductor_SMD:L_12x12mm	Resonant Inductor	15.0 uH, 20A saturation	Series resonant inductor for LLC tank
Cr	Capacitor_THT:C_Rect_P15.00mm	Resonant Capacitor	169 nF, 630V C0G/Polypropylene	High-dV/dt resonant tank capacitor
T1	Transformer_THT:Transformer_37x44	Planar EILP38 Transformer	Np:Ns=1:8, Lm=75uH, 500W	High-frequency isolation transformer (100kHz)
C201	Capacitor_SMD:C_0805	DC Bus Decoupling Cap	1.0 uF, 100V X7R	High-frequency ceramic local bypass for Leg A
C202	Capacitor_SMD:C_0805	DC Bus Decoupling Cap	1.0 uF, 100V X7R	High-frequency ceramic local bypass for Leg B

Sheet 3 (Page 4): 03_Secondary_Rectifier_DCLink — Secondary Rectification & 400V DC Link

Rectifies the 100kHz resonant transformer output to create a stabilized 380V-420V DC link. Features 4x C3D04060A SiC Schottky diodes (zero reverse recovery), RC snubbers, a 4x film capacitor DC link bank (18.8uF), active bleeders, and a precision telemetry divider.

Key Engineering Equations & Altitude Deratings:

* Rectified DC Bus: $V_{DC(bus)} = 2 * (N_s / N_p) * V_{PV} * M_{LLC} = 2 * 8 * 40V * 0.625 = 400.0\text{ V}$
* Film Cap Bank C: $C_{DC(total)} = C_{dc1} + C_{dc2} + C_{dc3} + C_{dc4} = 4 * 4.7\text{ uF} = 18.8\text{ uF } 630V$ (Polypropylene Film)
* Bleeder Discharge: $\tau = R_{bleed} * C_{total} = (3 * 100\text{ kOhm}) * 18.8\text{ uF} = 5.64\text{ s}$ -> Discharges < 50V safety limit in 11.7 s
* Divider Ratio & ADC: $K_{div} = 4.99\text{ kOhm} / (2 * 499\text{ kOhm} + 4.99\text{ kOhm}) = 1 / 201.0$ -> $V_{FB} = 400.0\text{ V} * (1 / 201.0) = 1.990\text{ V}$ to ADC

Complete Component Bill of Materials & Electrical Ratings:

Ref	Footprint / Library Symbol	Description	Electrical Ratings & Values	Circuit Function & Role
D301	Package_TO_SOT_THT:TO-220-2	SiC Schottky Diode	600V, 4A, Qrr=0 nC, Vf=1.5V	Full-bridge secondary rectifier high-side 1
D302	Package_TO_SOT_THT:TO-220-2	SiC Schottky Diode	600V, 4A, Qrr=0 nC, Vf=1.5V	Full-bridge secondary rectifier low-side 1
D303	Package_TO_SOT_THT:TO-220-2	SiC Schottky Diode	600V, 4A, Qrr=0 nC, Vf=1.5V	Full-bridge secondary rectifier high-side 2
D304	Package_TO_SOT_THT:TO-220-2	SiC Schottky Diode	600V, 4A, Qrr=0 nC, Vf=1.5V	Full-bridge secondary rectifier low-side 2
R_snub	Resistor_SMD:R_2512	Snubber Resistor	22 Ohm, 2W pulse rated	Damps high-frequency secondary ringing
C_snub	Capacitor_THT:C_Rect_P15.00mm	Snubber Capacitor	1.0 nF, 1000V C0G	Absorbs secondary switching transients
Cdc1	Capacitor_THT:C_Rect_P15.00mm	Film DC Link Cap	4.7 uF, 630V Polypropylene	High-reliability DC link capacitor 1
Cdc2	Capacitor_THT:C_Rect_P15.00mm	Film DC Link Cap	4.7 uF, 630V Polypropylene	High-reliability DC link capacitor 2
Cdc3	Capacitor_THT:C_Rect_P15.00mm	Film DC Link Cap	4.7 uF, 630V Polypropylene	High-reliability DC link capacitor 3
Cdc4	Capacitor_THT:C_Rect_P15.00mm	Film DC Link Cap	4.7 uF, 630V Polypropylene	High-reliability DC link capacitor 4
R301	Resistor_THT:R_Axial_P15.24mm	Bleeder Resistor	100 kOhm, 2W metal film	Active safety bleeder resistor 1
R302	Resistor_THT:R_Axial_P15.24mm	Bleeder Resistor	100 kOhm, 2W metal film	Active safety bleeder resistor 2
R303	Resistor_THT:R_Axial_P15.24mm	Bleeder Resistor	100 kOhm, 2W metal film	Active safety bleeder resistor 3
R_div1	Resistor_THT:R_Axial_P15.24mm	HV Divider Resistor	499 kOhm, 0.5W 0.1% tol	High-voltage divider top leg 1
R_div2	Resistor_THT:R_Axial_P15.24mm	HV Divider Resistor	499 kOhm, 0.5W 0.1% tol	High-voltage divider top leg 2
R_div3	Resistor_THT:R_Axial_P15.24mm	Precision Sense Resistor	4.99 kOhm, 0.125W 0.1% tol	Low-voltage sense leg to ADC
C_filt	Capacitor_SMD:C_0805	ADC Anti-Aliasing Cap	10 nF, 50V C0G	Low-pass filter for ADC voltage feedback

Sheet 4 (Page 5): 04_SiC_Inverter_LCL_Grid — SiC H-Bridge, LCL Filter & Grid Connection

Converts 400V DC into clean 230V RMS 50Hz AC grid power with <3% THD. Utilizes 4x Wolfspeed C3M0075120K 1200V SiC MOSFETs in 4-lead Kelvin source packages, 2x UCC21520 gate drivers, a damped LCL filter, dual redundant disconnect relays, and a current feedback loop.

Key Engineering Equations & Altitude Deratings:

* Grid AC Voltage: $V_{grid(rms)} = 230\text{ V AC } +/-\ 10\%$, $V_{grid(peak)} = 230 * \text{sqrt}(2) = \mathbf{325.27\text{ V}}$ (< 400 V DC bus)
* LCL Resonance: $f_{res} = 1 / (2 * \pi) * \text{sqrt}((L_{401} + L_{402}) / (L_{401} * L_{402} * C_f)) = 1 / (2 * \pi) * \text{sqrt}((1.8\text{ mH}) / (0.72\text{ mH}^2 * 2.2\text{ uF})) = \mathbf{4.96\text{ kHz}}$
* Damping Resistor: $R_d = 1 / (3 * 2 * \pi * f_{res} * C_f) = 1 / (3 * 2 * \pi * 4960 * 2.2\text{ uF}) = \mathbf{4.86\text{ Ohm approx } 4.7\text{ Ohm (5W non-inductive)}}$
* Grid Current Shunt: $V_{shunt} = I_{grid(peak)} * R_{sh4} * G_{U403} = 3.07\text{ A} * 10.0\text{ mOhm} * 20\text{ V/V} = \mathbf{0.614\text{ V}}$ (Instantaneous to ADC)

Complete Component Bill of Materials & Electrical Ratings:

Ref	Footprint / Library Symbol	Description	Electrical Ratings & Values	Circuit Function & Role
Q401	Package_TO_SOT_THT:TO-247-4	SiC MOSFET (Kelvin Source)	1200V, 30A, Rds=75mOhm	Inverter Leg A High-Side SiC Switch
Q402	Package_TO_SOT_THT:TO-247-4	SiC MOSFET (Kelvin Source)	1200V, 30A, Rds=75mOhm	Inverter Leg A Low-Side SiC Switch
Q403	Package_TO_SOT_THT:TO-247-4	SiC MOSFET (Kelvin Source)	1200V, 30A, Rds=75mOhm	Inverter Leg B High-Side SiC Switch
Q404	Package_TO_SOT_THT:TO-247-4	SiC MOSFET (Kelvin Source)	1200V, 30A, Rds=75mOhm	Inverter Leg B Low-Side SiC Switch
Rg401-404	Resistor_SMD:R_0805	SiC Gate Resistors (4x)	10 Ohm, 0.25W 1% tol	Gate turn-on damping resistors for Q401-Q404
U401	Package_SO:SOIC-16W	Isolated Dual Gate Driver	5.7 kVrms isolation, 4A/6A drive	Reinforced driver for Inverter Leg A
U402	Package_SO:SOIC-16W	Isolated Dual Gate Driver	5.7 kVrms isolation, 4A/6A drive	Reinforced driver for Inverter Leg B
L401	Inductor_SMD:L_12x12mm	Inverter Side Inductor	1.2 mH, 10A saturation	Inverter bridge high-frequency filter inductor
L402	Inductor_SMD:L_12x12mm	Grid Side Inductor	0.6 mH, 10A saturation	Grid side filter inductor
Cf	Capacitor_THT:C_Rect_P15.00mm	LCL Filter Capacitor	2.2 uF, 450V AC/630V DC	Filter shunt capacitance for LCL
Rd	Resistor_SMD:R_2512	LCL Damping Resistor	4.7 Ohm, 5W non-inductive	Passive damping resistor avoiding resonance
L403	Inductor_SMD:L_12x12mm	Neutral Return Choke	0.6 mH, 10A saturation	Differential-mode neutral line filter inductor
K401A	Relay_THT:Relay_SPST_Omron	Grid Disconnect Relay A	16A 250V AC, 12V coil	Primary AC Line disconnect relay
K401B	Relay_THT:Relay_SPST_Omron	Grid Disconnect Relay B	16A 250V AC, 12V coil	Neutral line disconnect relay
Q405	Package_TO_SOT_SMD:SOT-23	Relay Coil Driver NMOS	30V, 5.7A, Rds=18mOhm	Low-side driver for relay coils
D405	Diode_SMD:D_SOD-123	Relay Coil Clamp Diode	100V, 300mA, trr=4ns	Flyback diode suppressing coil kick
R_sh4	Resistor_SMD:R_2512	AC Current Shunt	10.0 mOhm, 3W, 0.5% tol	Inverter AC output current sensing
U403	Package_SO:SOIC-8	Current Sense Amplifier	Gain=20 V/V, 80V CM rejection	AC line current monitor for MPPT/anti-islanding
J401	TerminalBlock_Phoenix_MKDS-1,5-3	Grid Output Connector	230V AC (L, N, PE), 16A	3-pin high-current grid interface terminal

Sheet 5 (Page 6): 05_Control_DualMCU_Sensors — Isolated Dual MCU & High-Altitude Sensors

Implements an isolated dual-controller architecture: TM4C1230C3PM (80MHz ARM Cortex-M4F) dedicated to 100kHz LLC MPPT and 50kHz grid inverter SPWM, isolated via ISO7762DW from an ESP32-S3 wireless SoC running MQTT telemetry and BME280 barometric monitoring.

Key Engineering Equations & Altitude Deratings:

* Barometric Formula: $P(h) = 101.325 * (1 - 0.0065*5000 / 288.15)^{5.2558} = 54.05 \text{ kPa}$ -> Derating Factor $k_d = 1.48x$
* SPWM Resolution: $N_steps = f_clock / f_sw = 80 \text{ MHz} / 50 \text{ kHz} = 1600 \text{ Counts}$ (10.6 bits effective PWM resolution)
* Isolated SPI Bus: $f_SPI = 10.0 \text{ Mbps}$ (ISO7762DW rated up to 100 Mbps, $t_prop = 10.7 \text{ ns}$)
* ADC Quantization: $LSB = V_ref / 2^{12} = 3.300 \text{ V} / 4096 = 0.8057 \text{ mV / count}$

Complete Component Bill of Materials & Electrical Ratings:

Ref	Footprint / Library Symbol	Description	Electrical Ratings & Values	Circuit Function & Role
U502	Package_QFP:LQFP-64	Primary Control MCU	80MHz ARM Cortex-M4F, 256KB	Core DSP for MPPT, LLC & Grid Inverter SPWM
Y501	Crystal:Crystal_SMD_3225-4Pin	Precision Crystal	16.000 MHz, 10ppm tol, 18pF	Master system clock reference for TM4C123
C501	Capacitor_SMD:C_0805	Crystal Load Capacitor	18 pF, 50V C0G	Crystal load capacitance 1
C502	Capacitor_SMD:C_0805	Crystal Load Capacitor	18 pF, 50V C0G	Crystal load capacitance 2
C503	Capacitor_SMD:C_1206	VDDC Core Decoupling	2.2 uF, 10V X7R low-ESR	Internal 1.2V LDO regulator stabilization
C509	Capacitor_SMD:C_0805	MCU VDD Bypass Cap	100 nF, 16V X7R	Local high-frequency bypass for TM4C123
C510	Capacitor_SMD:C_0805	MCU VDD Bypass Cap	100 nF, 16V X7R	Local high-frequency bypass for TM4C123
U501	Package_SO:SOIC-16W	Reinforced Digital Isolator	5.0 kVrms isolation, 100Mbps	Isolates SPI & status buses between MCUs
C511	Capacitor_SMD:C_0805	Isolator Decoupling A	100 nF, 50V X7R	Side 1 local supply bypass for ISO7762
C513	Capacitor_SMD:C_0805	Isolator Decoupling B	100 nF, 50V X7R	Side 2 local supply bypass for ISO7762
U503	RF_Module:ESP32-S3-WROOM-1	Secondary Comms MCU	Dual-core 240MHz, Wi-Fi/BLE	Handles telemetry, cloud logging & OTA
U504	Package_LGA:Bosch_LGA-8	HAA Environmental Sensor	-40 to +85C, 30-110 kPa, I2C	Real-time barometric & temp monitor
R501	Resistor_SMD:R_0805	I2C Pull-Up Resistor	4.7 kOhm, 0.125W 1% tol	I2C SDA pull-up to +3V3_SEC
R502	Resistor_SMD:R_0805	I2C Pull-Up Resistor	4.7 kOhm, 0.125W 1% tol	I2C SCL pull-up to +3V3_SEC

Sheet 6 (Page 7): 06_Aux_Power_Supply — Auxiliary Multi-Rail High-Voltage Flyback

Derives all internal voltages directly from the 20V-60V PV input. Driven by a UCC3800 current-mode controller and a custom 3-winding flyback transformer T601, generating +15V_PRI (GaN drivers), +15V_SEC (SiC drivers), and regulated +5V / +3.3V rails for both domains.

Key Engineering Equations & Altitude Deratings:

* Flyback Oscillator: $f_{osc} = 1.5 / (R_{t601} * C_{t601}) = 1.5 / (15\text{ k}\Omega * 1.0\text{ nF}) = \mathbf{100.0\text{ kHz}}$
* Peak Primary Current: $I_{pk} = (2 * P_{aux}) / (\eta * V_{in(min)} * D_{max}) = (2 * 15W) / (0.80 * 20V * 0.45) = 4.17\text{ A} \rightarrow R_{603} = 1.0V / 4.17A = \mathbf{0.22\text{ }\Omega (2W)}$
* RCD Snubber Clamp: $V_{clamp} = V_{in(max)} + (N_p / N_s) * (V_{out} + V_d) = 60V + 1.33 * 15.7V = \mathbf{80.9\text{ V}}$ (Snubs primary drain spikes)
* Turns Ratios (T601): $N_p : N_{s1} : N_{s2} = \mathbf{20 : 15 : 15}$ (Generates +15V_PRI and +15V_SEC isolated auxiliary rails)

Complete Component Bill of Materials & Electrical Ratings:

Ref	Footprint / Library Symbol	Description	Electrical Ratings & Values	Circuit Function & Role
U601	Package_SO:SOIC-8 (UCC3800)	Current-Mode PWM Controller	BiCMOS, ultra-low start current	Auxiliary flyback controller (100kHz)
R601	Resistor_THT:R_Axial_P15.24mm	HV Start-Up Resistor	100 k Ω , 2W metal film	Trickle charge startup resistor for UCC3800
C_boot	Capacitor_THT:CP_Radial_D12.5mm	VCC Bootstrap Capacitor	47 uF, 35V low-ESR, 105C	Provides UCC3800 operating current at startup
Rt601	Resistor_SMD:R_0805	Timing Resistor	15.0 k Ω m, 0.125W 1% tol	Sets UCC3800 oscillator frequency to 100kHz
Ct601	Capacitor_SMD:C_0805	Timing Capacitor	1.0 nF, 50V C0G	Sets UCC3800 oscillator frequency to 100kHz
Q601	Package_TO_SOT_THT:TO-220-3	Flyback Power MOSFET	100V, 33A, Rds=44m Ω m	Primary flyback power switch
Rg601	Resistor_SMD:R_0805	MOSFET Gate Resistor	10 Ω m, 0.125W	Damps gate switching transients for Q601
R603	Resistor_SMD:R_2512	Current Sense Resistor	0.22 Ω m, 2W 1% tol	Peak current sensing resistor for PWM controller
D_clamp	Diode_SMD:D_SOD-123	RCD Clamp Diode	100V, 300mA, trr=4ns	RCD snubber fast clamping diode
R_clamp	Resistor_THT:R_Axial_P15.24mm	RCD Clamp Resistor	4.7 k Ω m, 2W metal film	Dissipates transformer leakage energy
C_clamp	Capacitor_THT:C_Rect_P15.00mm	RCD Clamp Capacitor	4.7 nF, 250V C0G	Snubs transformer primary drain spikes
T601	Transformer_Toroid_Tapped_D14.0mm	Flyback Transformer	Np:Ns1:Ns2=20:15:15, 15W	3-winding multi-rail auxiliary transformer
D601-602	Diode_SMD:D_SMA (SS34)	Schottky Rectifiers (2x)	40V, 3A, Vf=0.50V	Rectifies Sec 1 (+15V_PRI) & Sec 2 (+15V_SEC)
C601-602	Capacitor_THT:CP_Radial_D12.5mm	Filter Capacitors (2x)	220 uF, 35V low-ESR, 105C	Output filter capacitors for +15V rails
U603/U604	Package_TO_SOT_SMD:SOT-223-3	Linear Regulators (LM7805)	+5V Out, 1.0A rated	Regulates +15V rails down to +5V_PRI & +5V_SEC
C603/C605	Capacitor_SMD / THT	Regulator Filter Caps	47 uF THT / 10 uF SMD	Filter capacitors for +5V_PRI & +5V_SEC
U602/U605	Package_TO_SOT_SMD:SOT-23-5	Low-Dropout LDOs (TLV75533)	+3.3V Out, 500mA, low-noise	Regulates +5V down to +3.3V rails for MCUs
C607/C608	Capacitor_SMD:C_1206	LDO Output Caps (2x)	4.7 uF, 10V X7R ceramic	Output stability capacitors for +3.3V rails
D603	Package_TO_SOT_SMD:SOT-23	Precision Shunt Regulator	2.495V Ref, 0.5% tol	Voltage feedback reference for aux regulation
R602/R605	Resistor_THT:R_Axial_P15.24mm	Feedback Divider Resistors	10 k Ω m / 2.5 k Ω m, 0.1%	Voltage divider setting auxiliary output
C606	Capacitor_SMD:C_1206	Feedback Stability Cap	100 nF, 25V X7R	Frequency compensation for TL431 loop